

MALL BILLING SYSTEM - VIVA QUESTIONS & ANSWERS

STRUCTURES & OBJECT CONCEPTS

Question	Answer
What is a structure in C++?	A structure groups related variables of different types under one name.
Difference between struct and class?	Struct members are public by default, class members are private.
Why use two structs here?	Product stores item details, CartItem stores product with quantity.
What is nested structure?	When one structure contains another structure as a member.

VECTORS & FUNCTIONS

Question	Answer
Why use vector instead of array?	Vector is dynamic and provides functions like push_back().
Why const vector& used?	To prevent copying and modification of the original vector.
Why pass parameters by reference?	To modify original data without copying.
Purpose of findProduct()?	Searches a product by code and returns its pointer.
Why return pointer?	Avoids copying and allows checking for nullptr.

FILE HANDLING & FORMATTING

Question	Answer
What is file handling?	Reading or writing data from/to files using streams.
Use of ofstream?	Used to write data into files.
What is setw() and setprecision()?	setw sets column width, setprecision controls decimal digits.
Why fixed used with setprecision?	To show fixed decimal format like 499.50.

CONTROL FLOW & LOGIC

Question	Answer
Why use switch-case?	To handle multiple menu options clearly.
Why do-while loop used?	Ensures menu runs at least once before exit.
What if invalid product code?	Program shows 'Product code not found' message.
What if quantity <= 0?	Shows 'Invalid quantity!' message.

DISCOUNT & BILL CALCULATION

Question	Answer
How total calculated?	Sum of (price × quantity) for all items.
When discount applies?	5% for totals ≥1500, 10% for ≥3000.
How final bill displayed?	Shows product details, subtotal, discount, and final total.